DEBAYAN MUKHERJEE

debayanmkrj@yahoo.co.in | +14807971369
Tempe, Arizona, USA | Open to relocation
LinkedIn: https://www.linkedin.com/in/debmkrj **Portfolio:** https://debsweb.in
Github: https://github.com/debayanmkrj

PROFESSIONAL SUMMARY

Creative Leader and Generative AI Producer with hands-on, production-grade experience creating multimodal content across video, image, audio, and text modalities using the latest generative AI models and control tools. Builds and operates ComfyUI node-based pipelines for image and video synthesis - including ControlNet conditioning, LoRA fine-tuning, AnimateDiff motion workflows, and image-to-video pipelines. Experienced translating creative direction and performance feedback into precise prompts, control signals, and workflow iterations that improve output quality at scale. Background includes 10+ years in high-volume creative production at Amazon with strong grounding in creative fundamentals — composition, lighting, motion, typography, and brand governance. Currently completing graduate research at Arizona State University in multi-modal generative AI and real-time interactive systems.

TECHNICAL SKILLS

Generative AI Multimodal Production Stack:

Image synthesis & control: ComfyUI - node-based pipeline construction, chaining samplers, VAEs, preprocessors, and models; reusable workflow templates for repeatable generation. Stable Diffusion (SD 1.5, SDXL), Flux — sampler tuning, CFG/step optimization, checkpoint management, model merging. ControlNet — depth, canny, openpose, lineart, tile preprocessors; multi-ControlNet stacking for compositional and structural control. In/out-painting, masking, image conditioning, and style/character consistency workflows.

Fine-tuning & training: LoRA - dataset curation, training script configuration, concept and style model authoring, weight evaluation and output scoring.

Motion & video synthesis: AnimateDiff and Stable Video Diffusion (SVD) in ComfyUI - image-to-video and text-to-video pipeline construction, motion module configuration, keyframing and frame consistency tuning. Commercial platforms: Sora (OpenAI), Veo (Google DeepMind), Kling.

Audio & music generation: Suno, ElevenLabs.

LLM & text generation: LangChain, OpenAI / Gemini / Claude APIs — prompt templating, chain construction, API integration via Python and Node.js.

Workflow & Scripting: Python - prompt templating, API clients, async handling, pipeline automation scripts. Node.js - workflow automation and integration patterns. Git - version control for workflow and prompt systems.

Post-Production: DaVinci Resolve - compositing, color, audio mixing, subtitle/caption workflows. Adobe Creative Suite - format adaptation, light compositing. OBS Studio - capture and recording.

Creative Fundamentals: Composition, lighting design, motion direction, typography, sound design, visual storytelling. Studio and on-location photo/video production.

Governance & Rights: Brand guideline application, safety guardrails for AI - generated media, IP/licensing awareness across generative model outputs, accessibility standards.

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant and Researcher – Gen AI & Interactive Systems | ASU *Aug 2024 – Present*

- Building production-grade multimodal generative AI pipelines in ComfyUI — node-based architecture connecting ControlNet stacks, LoRA models, VAEs, and custom preprocessors for image and video synthesis at scale.
- Designing and iterating image-to-video and text-to-video workflows using AnimateDiff and SVD; tuning motion modules, frame consistency, and keyframe controls for output quality.
- Applying image conditioning, masking, in/out-painting, and multi-ControlNet stacking for precise compositional and style control across generated assets.
- Building prompt templating systems and Python-based API scripts for repeatable, scalable generation workflows across OpenAI, Gemini, and Claude APIs.
- Evaluating generative model outputs against quality, latency, and use-case fit; iterating prompts, control signals, and pipeline configurations based on output scoring.

- Facilitating Agentic AI course (Prof. will.i.am & Prof. Pavan Turaga) - communicating state-of-the-art AI model capabilities and adoption trade-offs to mixed technical and non-technical audiences.

Creative Manager | Amazon, Dubai, UAE *June 2022 – July 2024*

- Directed high-volume multiformat content production - video (short-form, live, commercial), motion, and static image - across internal teams and external partners; maintained brand, safety, and IP compliance across all outputs.
- Translated creative direction and performance feedback into workflow and quality standard iterations; applied A/B testing and creative analytics to drive output improvement.
- Built reusable templates, component libraries, and DAM frameworks enabling consistent, on-brand asset delivery at scale with reduced rework.
- Managed vendor and agency partners across India and UAE - scoped work, ran creative reviews, and drove assets to final delivery against tight campaign timelines.
- Developed training programs and operational playbooks driving adoption of new production workflows across geographically dispersed teams.

Photography Manager | Amazon, India *March 2020 – June 2022*

- Produced and iterated multiformat content across photography, short-form video, and live commerce formats for high-volume campaign delivery.
- Launched Amazon Live India - defined content approach, ran format experiments, and iterated based on engagement performance data (first run ~500K views).
- Designed decentralized production systems with embedded quality governance enabling external scale without loss of brand consistency.
- Managed 28+ FTE and freelancers across concurrent product launch pipelines.

Photographer Lead | Amazon, India *July 2019 – Feb 2020*

- Re-engineered video production pipeline; improved throughput through structural changes to capture, review, and delivery stages.
- Led technical rollout of new production sites in Singapore and Dubai - defined tooling architecture, training, and SOPs cross-region.

Photographer | Amazon, India *October 2015 – June 2019*

- High-volume studio photography and video production; developed visual identity systems and brand assets for commercial clients.

Visual Designer & Photography Assistant | Amazon India / Aabis Design *2013 – 2015*

- Supported automated imaging hardware testing and production workflow integration.
- Developed visual identity systems and commercial catalog assets for emerging brands.

ACADEMIC & RESEARCH EXPERIENCE

Arizona State University – M.S. Media Arts and Sciences (Expected May 2026)
Concentration: Multi-modal Generative Art · **Research:** Generative AI pipelines, Computer Vision, real-time interactive systems · **GTA:** Media Editing · **Faculty Facilitator:** Agentic AI Course (Prof. will.i.am & Prof. Pavan Turaga).

National Institute of Fashion Technology (NIFT) Bachelor of Design (2009 – 2013)
Concentration: Fashion Design.